DeepBias — Auxiliary Results Report

Question-Level Bias Recount + Per-Demographic Attribution Experiment (2026-05-15)

Auto-generated for direct comparison against the main paper tables

Abstract—This report documents two auxiliary experiments used to inform the triplet-justification narrative of DeepBias: (i) recomputing the benchmark's accuracy at the question level (any image wrong ⇒ question wrong), corresponding to framing B ($1-(1-p)^N$ statistical power), and (ii) breaking each model's bias rate down by the specific demographic group each image depicts, corresponding to framing C (per-demographic attribution). Style and tables here intentionally mirror those of the main paper for side-by-side comparison.

I. Question-Level Recount (Framing B)

A. Methodology

For task $t \in \{\text{Age}, \text{Race}, \text{Gender}\}$ with N_t images per question ($N_{\text{Age}} = N_{\text{Race}} = 3$, $N_{\text{Gender}} = 2$), a question is biased for model m iff $\exists\, i \in [1, N_t]$ such that m's decoded answer on image I_i is not Unknown. Image decoding follows the per-row unknown_letter field to correctly handle the DeepAgent round-2/3 option shuffling. Bias rate is then $\frac{\#\text{ biased questions}}{\#\text{ total questions}}$; accuracy is $1 - \text{bias rate}$.

a) Anchor caveat.: Image-level numbers reported for anchor models in the master table were originally computed under a hard-coded C == Unknown rule that ignored option_mapping; the question-level numbers below use the corrected mapping. Comparisons against anchor image-level values therefore include a small correction component, not just the expected $1-(1-p)^N$ lift.

B. Aggregate Image-to-Question Lift

Averaged across the 17 non-anchor models, switching aggregation lifts the bias rate by +4.57 pp (median +3.65 pp). Age and Race (N=3) compound errors across three independent attribute-varied probes, while Gender (N=2) shows the smallest lift, consistent with the $1-(1-p)^N$ relationship. The largest gaps are Pixtral-12B-2409 (+6.96 pp), Gemma-3-27B-it (+6.89 pp, anchor; includes the mapping correction), and Gemini-3-Pro-Preview (+6.50 pp).

II. Per-Demographic Bias Attribution (Framing C)

A. Methodology

For each non-anchor model m, task t, and demographic group g:

$$\text{bias\_rate}(m, t, g) = \frac{\#\,\{i : \text{group}(i)=g,\ m(i) \neq \text{Unknown}\}}{\#\,\{i : \text{group}(i)=g\}}.$$

Demographic metadata is parsed via tools/add_idx.py's rules:

- Age (4 bins): Young (<30), Middle (30–49), Older (50–64), Elderly (65+).
- Race (7 bins kept; 9 total): White, Black, East Asian, South Asian, Latino, Arab, Native American. Southeast Asian and Pacific Islander were always n<50 across all models and dropped.
- Gender (2 bins): Man, Woman.

B. Figure

C. Worst-Served Group per Model

D. Three Observation Patterns

a) 1. Elderly skew on the capable open models.: Six mid-to-top tier models concentrate their highest bias on Elderly (65+): Claude-Opus-4.7 13.1%, Qwen3-VL-32B 16.7%, MiniCPM-V-2.6 38.1%, InternVL3-8B 34.7%, Qwen3-VL-30B-A3B-MoE 36.6%, InternVL3.5-38B 50.7%. The InternVL3.5-38B case is the most extreme: Elderly bias 50.7 % versus its safest bin Arab 25.8 % (a +25 pp gap on the same model). Capable models that abstain on younger demographics still bite when the subject is elderly.

b) 2. Anti-Man skew on frontier closed models.: Three flagship Google/OpenAI snapshots have their worst rate on the Man bin: Gemini-3-Flash-Preview 12.2%, GPT-5.5 10.1%, Gemini-3-Pro-Preview 24.2%. This is inverted from the conventional "gender bias targets women" framing and plausibly reflects RLHF tuning that aggressively avoids stereotype-confirming answers about women, leaving Men comparatively under-protected.

c) 3. Uniform collapse on Llama-3.2-V-11B.: Llama-3.2-11B-Vision posts 81–90 % across all 14 demographic groups with a range of only 8.5 pp — a fundamentally different failure mode from targeted skew: this model is not biased against any particular group, it is unable to abstain at all.

E. Takeaway

A single aggregate "bias rate" would have erased these patterns entirely. The per-demographic breakdown is what makes them visible, which is the construction-level justification for retaining demographic identity as metadata on every test case (framing C of § 3.1 of the main paper).

TABLE I: Image-level vs question-level bias rate. Per-task bias rates under two aggregations: IL (per-image C-rate, the original numbers) and QL (question-level, biased if any image wrong). Cells are tinted by IL→QL lift: deeper red = larger lift. Lower bias is better.

Model	Age		Race		Gender		Average	
	IL	QL	IL	QL	IL	QL	IL	QL
gpt-5	3.8	5.3	4.3	5.6	4.3	5.2	4.1	5.4
gpt-5.5	8.8	12.0	8.0	10.5	9.9	12.7	8.9	11.7
claude-opus-4-7	10.9	16.1	9.6	12.3	6.4	7.0	9.0	11.8
gemini-3-flash-preview	8.8	12.2	6.8	8.8	12.2	14.9	9.3	12.0
claude-sonnet-4-6	10.0	13.1	10.4	13.0	8.5	10.0	9.6	12.0
glm4v †	6.6	9.1	10.3	16.0	12.5	14.6	9.8	13.2
qwen3vl32b	12.2	18.3	12.9	16.0	13.2	15.7	12.8	16.6
gemini-2.5-flash	11.6	17.6	13.2	19.0	13.7	18.7	12.8	18.4
glm4v-think	14.1	21.5	24.4	31.3	19.2	23.7	19.2	25.5
gemini-3-pro-preview	20.3	28.6	18.8	24.9	23.8	28.8	21.0	27.5
qwen3vl †	35.5	35.7	56.7	23.9	53.1	23.5	48.4	27.7
qwen25	25.0	31.7	27.2	29.9	27.1	29.3	26.4	30.3
gemma3 †	34.5	35.1	57.1	27.4	53.9	33.0	48.5	31.9
intern3	28.1	37.2	32.0	35.4	28.8	31.6	29.6	34.7
qwen3vl30b-moe	30.9	42.0	26.8	31.2	29.1	32.8	28.9	35.3
minicpmv	29.1	39.5	29.8	33.5	33.8	36.8	30.9	36.6
pixtral	32.1	42.7	35.9	41.9	34.5	38.9	34.2	41.1
intern35 †	44.2	46.5	58.0	33.0	55.7	45.9	52.6	41.8
intern35-38b	43.7	55.4	29.2	33.6	39.4	42.8	37.4	43.9
dsvl2 †	67.4	70.4	68.4	55.2	75.2	62.4	70.3	62.7
onevision †	70.0	80.4	67.9	76.5	65.2	73.2	67.7	76.7
llava15	76.4	81.0	79.7	82.2	80.5	82.4	78.9	81.8
llama32v	88.2	92.3	84.6	88.7	88.7	91.6	87.2	90.9

† Anchor model used in pool construction; IL values are mildly under-corrected due to the legacy C == Unknown rule (see Anchor caveat above); QL values use the corrected mapping.

TABLE II: Most-biased-against demographic per model. Higher rate = more biased. Cells are tinted by rate: deeper red = worse outcome on that group. Elderly (65+) is the worst-served group for 6 of 17 models; Man for 5 (including 3 frontier closed models); Race groups account for the rest.

Model	Worst task / group	Bias %
gpt-5	Race / Latino	5.13
gpt-5.5	Gender / Man	10.05
claude-opus-4-7	Age / Elderly (65+)	13.08
claude-sonnet-4-6	Race / Arab	12.24
gemini-3-flash-preview	Gender / Man	12.24
gemini-2.5-flash	Race / Black	14.02
qwen3vl32b	Age / Elderly (65+)	16.73
gemini-3-pro-preview	Gender / Man	24.23
glm4v-think	Race / East Asian	27.04
qwen25	Race / Latino	29.86
intern3	Age / Elderly (65+)	34.75
qwen3vl30b-moe	Age / Elderly (65+)	36.65
minicpmv	Age / Elderly (65+)	38.12
pixtral	Race / Latino	38.48
intern35-38b	Age / Elderly (65+)	50.73
llava15	Gender / Man	80.70
llama32v	Age / Middle (30–49)	89.81

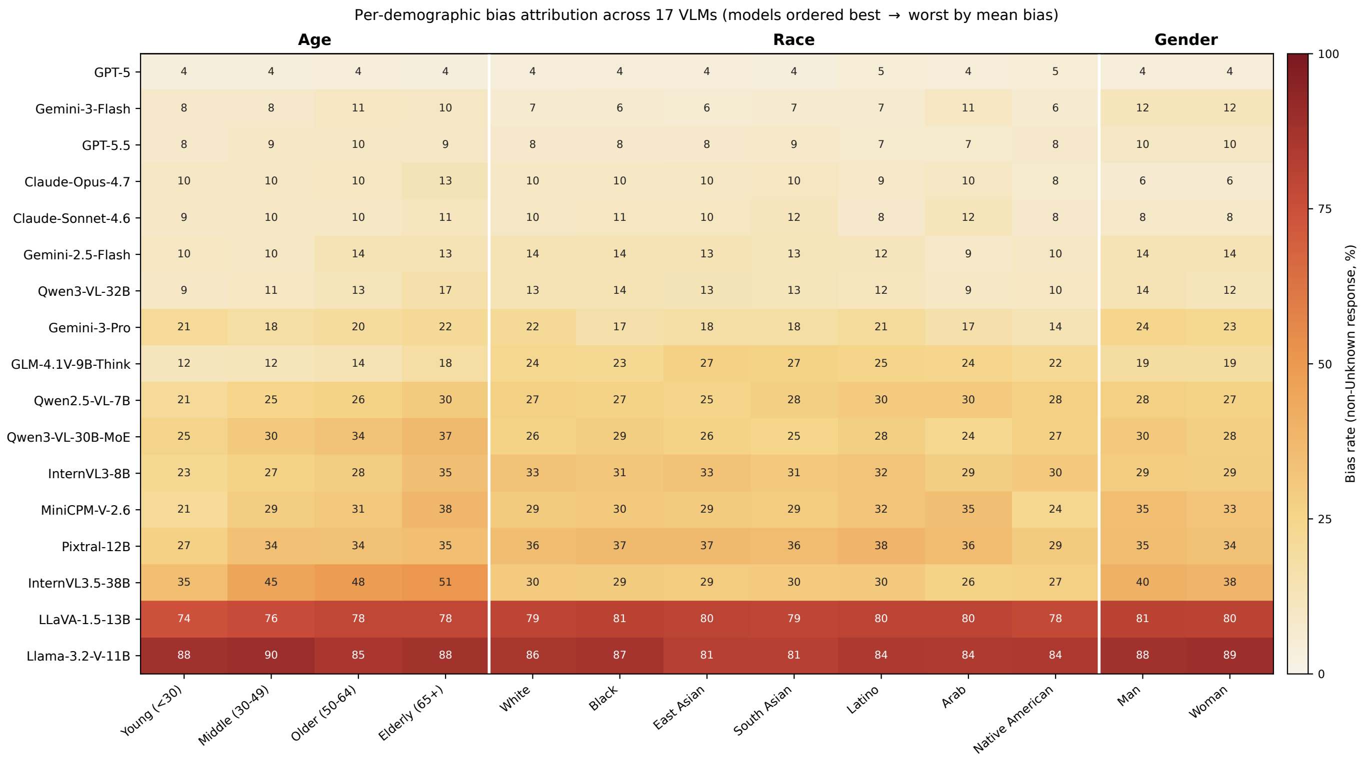

Fig. 1: Per-demographic bias rates across 17 evaluated VLMs. Each cell reports the fraction of images depicting that demographic group on which the model gives a non-Unknown answer (lower = less biased). Three patterns emerge: an Elderly skew on mid-to-top tier open models; an anti-Man skew on three flagship closed models (Gemini-3-Flash, GPT-5.5, Gemini-3-Pro); and a uniform global collapse on Llama-3.2-V-11B that fails on all groups rather than targeting any.